

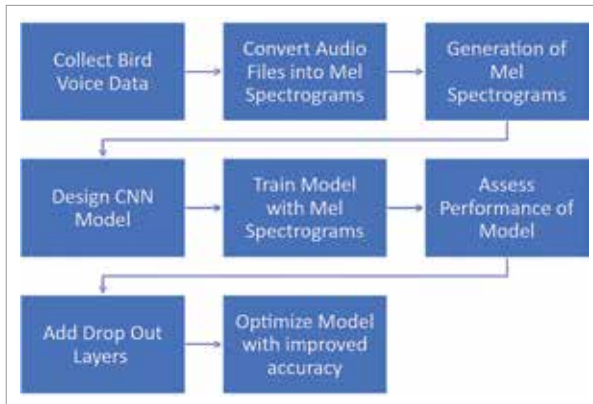


Figure 1: Bird voice recognition process

can be heard in natural habitats and can obscure bird cries.

- **Recording quality:** Quality of the recorded bird cries may be impacted by differences in the tools and methods used for recording.
- **Limited data:** Data is scarce and imbalanced for many bird species, particularly the uncommon ones.

Methods used for bird voice recognition

Various techniques are used to recognise the cries of different species of birds and analyse them:

- Audio recordings and software tools to identify bird species by their calls
- Spectrogram analysis techniques
- Customised applications that monitor bird populations and conservation efforts
- Avian studies on biodiverse areas combining traditional methods with bird voice recognition
- Automated recording units that are deployed to record bird calls continuously
- Machine learning algorithms to analyse recordings to identify species and call frequencies
- Citizen science programmes that engage the community in data collection

Developing a bird voice recognition model

The creation of a bird voice recognition

model involves collecting audio recordings of birds, converting these into Mel spectrograms, training a CNN model to classify the Mel spectrograms, and evaluating and refining the model's performance.

Figure 1 depicts the process of creating a bird voice model.

▪ **Collecting bird voice data:**

In this initial step, recordings of various bird calls from numerous species and locations are gathered. The data selection process might involve focusing on birds found in a particular region

▪ **Generating Mel spectrograms for bird voices:**

The audio files of bird voices are then converted into Mel spectrograms. The latter is a visual representation of the spectrum of frequencies of a signal that varies with time. The frequencies are scaled according to the Mel scale. This scale is designed for humans to perceive the pitch of sounds, which is useful in the identification of bird voices.

▪ **Designing a CNN model:**

A convolutional neural network (CNN) is a type of machine learning model that is particularly suited for image recognition tasks. A CNN model that is appropriate for the task of identifying bird voices from the Mel spectrograms must be designed.

▪ **Training the model with Mel spectrograms:**

The CNN model must be trained using the generated Mel spectrograms. During the training process, the model is input with numerous examples of Mel spectrograms, along with corresponding information about the bird calls. The model learns to recognise patterns in the Mel spectrograms that differentiate the

calls of various bird species.

- **Assessing the initial performance of the trained model:** Next, the performance of the trained model must be assessed to determine how well it can correctly classify bird voices from the new Mel spectrograms it has not been exposed to during training.
- **Improving model accuracy:** The addition of a dropout layer will help in improving model accuracy.
- **Optimised model with improved accuracy:** The final step is an optimised model, which has significantly improved accuracy when it comes to classifying bird voices.

You can use the following steps to generate a functioning Mel spectrogram:

- Load audio file using librosa.
 - Generate the Mel spectrogram from the audio data.
 - Transform the Mel spectrogram into a decibel (log) scale.
 - Normalise the spectrogram values to a range of [0, 1].
 - Plot the normalised Mel spectrogram and save it as an image file in the specified directory.
- Tentative code snippets are:

```
def savepicture(audio_path,dirval,file):
    y, sr = librosa.load(audio_path,
sr=None)

# Compute Mel spectrogram
S = librosa.feature.
melspectrogram(y=y, sr=sr)

# Convert to decibels (log scale)
S_db = librosa.power_to_db(S,
ref=np.max)

# Normalize to range [0, 1]
S_normalized = (S_db - np.min(S_db))
/ (np.max(S_db) - np.min(S_db))

# Plot Mel spectrogram
plt.figure(figsize=(10, 4))
plt.axis("off")
librosa.display.specshow(S_
```